No-Go Theorems for One-Step Lyapunov Potentials for the $3x + 1$ Map

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Abstract

A natural strategy for the $3x + 1$ (Collatz) problem is to exhibit a potential $\Phi(x) = \log_2 x + g(x)$ that is nonincreasing along the accelerated odd map $f(x) = (3x+1)/2^{v_2(3x+1)}$, with a correction g computed from simple features of x. It is folklore that no such monotone invariant is known; we prove that for broad natural classes of corrections none can exist. The master mechanism is *shadowing an expanding rational cycle*: the cycle $-5 \mapsto -7 \mapsto -5$ has multiplier $3^2/2^3 > 1$, and its positive-integer shadows $x \equiv -5 \pmod{2^N}$ traverse its coordinate itinerary while gaining value, so any correction that cannot separate x from $f^2(x) = \frac{9}{8}x + \frac{5}{8}$ is refuted by two integers — the smallest certificate being $1275 \mapsto 1913 \mapsto 1435$. This excludes all corrections $g\big(x \bmod 2^m,\ \tau(x),\ \mathrm{len}(x),\ \lambda_1(x), \ldots, \lambda_d(x)\big)$, where τ is the trailing-ones count and the λ_i are suffix-pattern detectors of arbitrary depth (Theorem A). Iterating the shadow and applying Dirichlet approximation to $\log_2 \frac{9}{8}$ extends the exclusion to corrections using any fixed number of leading binary digits (Theorem B); a Mersenne burn argument excludes all bounded corrections of arbitrary dependence (Theorem C). The detectors are not ad hoc: they read a canonical structure, an exact ancestry tower $w_d(M) = \big(2^{M+2d} - 2^{2d+1} + 3^d\big)/3^d$ with $f^d(w_d(M)) = 2^M - 1$, supported on $M \equiv 1 \pmod{2 \cdot 3^{d-1}}$ with a carry-free periodic binary normal form (Theorem D). The remaining natural coordinate, the size of x read to a fixed precision $\lfloor 2^j \log_2 x \rfloor$, falls as well: closed certificates assembled from several shadows of rational cycles exclude it at every precision j, jointly with residues modulo an arbitrary N and all detectors (Theorem E), and the least number of witnesses needed at precision 2^j is a semiconvergent denominator of $\log_2 3$ $(2, 2, 2, 7, 7, 12, 12, 12, 53, 665, \ldots)$. The exclusions persist for potentials that are nonincreasing only along f^k. The failures are rare but not too rare: a correction that reads m bits of x can fail on a set of density about 2^{-cm} with $c = 1 - H(1/\log_2 3) = 0.0500\ldots$, the large-deviation exponent for an orbit not descending within m parity steps, and the shadows of all expanding cycles of period $m+1$ show that no such correction fails less often (Theorem G). The same cycle also blocks a second method: on the Mersenne spine $2^n - 1 \mapsto (3^n - 1)/2$, no finite set of residue-class certificates on n can force a positive block score, because a 2-adic index n^* with $3^{n^*+3} = -367$ sends the orbit into $-5 \mapsto -7$ (Theorem F).

1 Introduction

For odd $x \in \mathbb{Z}_{>0}$ let

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}},$$

*Repository: `github.com/docbgm2002/collatz-things`. Every numbered claim is machine-verified in exact integer or rational arithmetic; see Section 12.

the accelerated (odd-to-odd) form of the Collatz map; the $3x+1$ conjecture asserts that every orbit reaches 1. Among the strategies proposed for the problem, one of the most natural is the *potential-function* (Lyapunov) method: find $\Phi : 2\mathbb{Z}+1 \to \mathbb{R}$ with $\Phi(f(x)) \le \Phi(x)$ for all odd $x > 1$, ideally of the coercive form $\Phi(x) = \log_2 x + g(x)$ with g a correction of lower order, so that monotonicity constrains orbit growth. Heuristics support the idea: a "random" odd step multiplies x by $3/2^e$ with e geometrically distributed of mean 2, so $\log_2 x$ decreases by $\log_2 \frac{4}{3}$ *on average* [6]. The difficulty, as surveys note, is that no correction making the decrease pointwise has ever been found [7]. To the best of our knowledge the literature contains no theorem *excluding* a concrete class of such potentials; the absence is reported as an empirical fact. This paper proves exclusions, and locates the precise boundary at which the exclusion program must stop.

Throughout, $\tau(x) = \mathrm{v}_2(x+1)$ is the number of trailing `1`s in the binary expansion of odd x (the *fuel*: Lemma 2.3 shows an odd step converts one unit of fuel into a multiplication by almost exactly $\frac{3}{2}$); $\mathrm{len}(x)$ is the bit length; $\mathrm{LD}_j(x) = \lfloor x/2^{\mathrm{len}(x)-j} \rfloor$ is the top j bits; and $\lambda_1, \lambda_2, \ldots$ are the suffix-pattern detectors of Definition 2.5, which read the run length of a canonical hierarchy of periodic binary patterns (levels 1 and 2 being $(\mathtt{10})^k\mathtt{1001}$ and $(\mathtt{111000})^k\mathtt{01}$). Our main impossibility results:

Theorem A (master no-go; Theorem 3.2). For every $m, d \ge 0$ and every function g, the potential $\log_2 x + g\big(x \bmod 2^m,\ \tau(x),\ \mathrm{len}(x),\ \lambda_1(x), \ldots, \lambda_d(x)\big)$ is not nonincreasing along f on odd $x > 1$.

Theorem B (leading digits; Theorem 4.1). For every $m, j, d \ge 0$ and every g, the potential $\log_2 x + g\big(x \bmod 2^m,\ \tau(x),\ \mathrm{LD}_j(x),\ \lambda_1(x), \ldots, \lambda_d(x)\big)$ is not nonincreasing along f on odd $x > 1$.

Theorem C (bounded corrections; Theorem 5.1). For every bounded $G : 2\mathbb{Z}+1 \to \mathbb{R}$, of arbitrary dependence on x, the potential $\log_2 x + G(x)$ is not nonincreasing along f on odd $x > 1$.

Theorem E (quantized logarithm; Theorem 7.6). For every $j, m, d \ge 0$, every $N \ge 1$ and every g, the potential $\log_2 x + g\big(\lfloor 2^j \log_2 x \rfloor,\ \tau(x),\ x \bmod N,\ \lambda_1(x), \ldots, \lambda_d(x)\big)$ is not nonincreasing along f on odd $x > 1$; nor is it nonincreasing along f^k for any fixed k.

Theorem A is proved by two integers per parameter choice: the map has the expanding rational cycle $-5 \mapsto -7 \mapsto -5$, of multiplier $3^2/2^3 = \frac{9}{8} > 1$ (expansion is exactly why it lives at negative integers), and its shadow at positive $x \equiv -5 \pmod{2^N}$ reproduces the cycle's coordinate itinerary while the value gains a factor $> \frac{9}{8}$ per period. Any correction blind to the difference between x and $f^2(x) = \frac{9}{8}x + \frac{5}{8}$ is refuted outright; the smallest certificate is $1275 \mapsto 1913 \mapsto 1435$. Theorem B iterates the shadow a times and chooses a, by Dirichlet approximation to the irrational $\log_2 \frac{9}{8}$, so that the accumulated gain sits next to an exact power of 2 and the leading digits return. Theorem C is a one-paragraph consequence of the Mersenne burn. Theorem E replaces the single shadow by a closed cycle of K independent witnesses, each shadowing one member of a rational cycle chosen to fit the precision, and the number K needed is exactly the least denominator at which $\lfloor K \log_2 3 \rfloor / K$ under-approximates $\log_2 3$ at least as well as $\lfloor 2^j \log_2 3 \rfloor / 2^j$ does (Section 7).

The failures produced by these theorems are explicit but sparse, and Section 8 asks how sparse they must be:

Theorem G (failure density; Theorem 8.2). Let δ_m be the infimum, over all g, of the lower density of odd x at which $\log_2 x + g(x \bmod 2^m)$ increases along f. Then $2^{-cm}/(16(m+2)^5) \le \delta_m \le 4 \cdot 2^{-cm}$ for $m \ge 12$, where $c = 1 - H(1/\log_2 3) = 0.0500\ldots$ and H is the binary entropy.

The lower bound sums the shadow mechanism over all expanding cycles of period $m+1$, whose number is governed by the entropy $H(1/\log_2 3)$; the upper bound is a potential that looks ahead along the orbit as far as m bits allow. The constant is the one that controls the density of integers with large stopping time, so a local correction can never do better than wait for the orbit to descend.

The detectors quantified over in Theorems A and B are given meaning by a structural result of independent interest:

Theorem D (Mersenne ancestry tower; Theorem 6.1). For $d \geq 1$ and $M \equiv 1 \pmod{2 \cdot 3^{d-1}}$, $M > 1$, the integer $w_d(M) = \big(2^{M+2d} - 2^{2d+1} + 3^d\big)/3^d$ is odd with $\tau = 1$, satisfies $3w_d(M) + 1 = 4\, w_{d-1}(M)$ and hence $f^d(w_d(M)) = 2^M - 1$ with every step of value ratio in $(\frac{3}{4}, 1)$; its binary expansion is exactly periodic (carry-free) of period $2 \cdot 3^{d-1}$; and $w_d(M) \equiv 1 - 2^{2d+1}3^{-d} \pmod{2^{M-1}}$, distinct levels separating 2-adically at bit $2\min(d, d') + 1$ exactly.

The tower is the complete uniform-e=2 ancestry of the Mersenne numbers — the slowest exact approach to maximal fuel — and λ_d detects membership in its level-d suffix pattern. The congruence domain is the order of 2 modulo 3^d, and the carry-free normal form is an instance of the lifting-the-exponent lemma.

Section 10 turns from potentials to congruence case analysis. After the burn, $2^n - 1$ reaches the repunit $a_n = (3^n - 1)/2$, and a natural attack on its stopping time scores the orbit of a_n block by block and splits on residues of n. Each such case is an exact finite computation (Proposition 10.4), but the method cannot cover all n:

Theorem F (Theorem 10.7). On $n = 64k + 17$ with k odd, the 2-adic index k^* defined by $3^{64k^*+20} = -367$ sends the orbit of a_n into the cycle $-7 \mapsto -5 \mapsto -7$, every block of which has negative score. Integers $k \equiv k^* \pmod{2^J}$ follow it for $\lfloor (J+2)/3 \rfloor$ blocks, and no finite set of certified residue classes covers all odd k.

Method and provenance

The constraint "Φ nonincreasing" restricted to any coordinate window is a difference-constraint system on the values of g; it is infeasible if and only if the coordinate graph of real orbit steps contains a directed cycle whose product of value ratios exceeds 1 (Section 9). Mining such cycles by Bellman–Ford over odd $x \leq 10^6$ produces startlingly short certificates — e.g. the pair $57 \mapsto 43$, $27 \mapsto 41$ already refutes all $g(x \bmod 16, \tau)$ — and inspection of the mined witnesses revealed that they shadow the -5 cycle, which led to the uniform theorems above. We record this provenance because the historical route of this project ran in the opposite direction: Theorems A–C were first proved for subclasses by designed families (a Mersenne "burn" that forces linear growth of g along a frozen coordinate line, against a "recharge" family that caps it), a method that yields an exact price list — each level of suffix detection buys a would-be potential exactly $\log_2 \frac{4}{3}$ of budget against an unbounded liability — but is dominated, as a source of impossibility statements, by the two-integer shadow certificates. The family analysis survives in the repository accompanying this paper and in the structural Theorem D.

The boundary

A coordinate defeats a single shadow only by separating x from $f^{2a}(x)$ for every good Dirichlet approximation a; the quantized logarithm $\lfloor 2^j \log_2 x \rfloor$ does so for $j \geq 3$, since one shadow period gains $\log_2 \frac{9}{8} > 2^{-j}$. Theorem E shows that this refuge is illusory: certificates built from several

witnesses reset their position in the cell at every junction, and exist at every precision. After Theorems A, B and E, no correction built from finitely much arithmetic information about x — its residue modulo any fixed integer, any fixed-depth suffix pattern, its size to any fixed precision — makes $\log_2 x$ monotone. Section 11 records what is left: the leading digits taken jointly with the quantized logarithm, and corrections that depend on the orbit rather than on x, for which the existence of a monotone potential is conjecture-adjacent.

Relation to the literature

The parity-vector structure underlying the heuristic goes back to Terras [13] and Everett [3]; density results culminate in [5] and, by different methods, in Tao's almost-all theorem [12]. Cycle exclusions rest on transcendence bounds [11, 10] and computation [2]. Backward (predecessor) structure of the Collatz graph is developed at the level of density and measure by Wirsching [14]; the tower of Theorem D is a distinguished explicit chain in that graph, and its closed form, congruence domain, and 2-adic freezing appear to be new. Forward identities of burn type, attached to the rational cycles at -1, -5, -17, appear in Andaloro [1] — the shadows of Section 3 are those identities viewed through the certificate lens. A search of the literature, including the annotated bibliographies [8, 9], located no prior impossibility theorems for concrete classes of one-step potentials; we would welcome references.

2 Preliminaries

Odd integers are written x; v_p is the p-adic valuation; $\tau(x) = \mathrm{v}_2(x+1)$; congruential inverses are taken in $(\mathbb{Z}/2^m\mathbb{Z})^\times$. Two standard facts are used.

Lemma 2.1 (LTE, special case)**.** *For even n, $\mathrm{v}_3(2^n-1) = 1+\mathrm{v}_3(n)$; in particular $\mathrm{v}_3\big(2^{2\cdot 3^{d-1}}-1\big) = d$.*

Lemma 2.2. *2 is a primitive root modulo 3^d for every $d \geq 1$; hence $\mathrm{ord}_{3^d}(2) = \varphi(3^d) = 2\cdot 3^{d-1}$.*

Proof. 2 generates $(\mathbb{Z}/9\mathbb{Z})^\times$ (its order divides 6 and is not $1, 2, 3$), and a primitive root mod 9 is a primitive root mod 3^d for all d. □

Lemma 2.3 (fuse)**.** *If $\tau(x) = t \geq 2$ then $\mathrm{v}_2(3x+1) = 1$, $f(x) = \frac{3x+1}{2}$, $\tau(f(x)) = t-1$, and $f(x)/x = \frac{3}{2} + \frac{1}{2x}$.*

Proof. Write $x+1 = 2^t u$, u odd. Then $3x+1 = 2\,(3\cdot 2^{t-1}u - 1)$ with odd bracket since $t \geq 2$, and $f(x)+1 = 3\cdot 2^{t-1}u$. □

Remark 2.4. The iterated form of Lemma 2.3 — that $2^t u - 1$ (u odd) iterates in t odd steps to $3^t u - 1$, the *burn* — is classical; it is recorded by Andaloro [1] (cf. the annotation in [9]).

Finally we define the detectors. For $d \geq 1$ write $\mathrm{ord}_d = 2\cdot 3^{d-1}$ and, for $M \equiv 1 \pmod{\mathrm{ord}_d}$, $M > 1$,

$$w_d(M) = \frac{2^{M+2d} - 2^{2d+1} + 3^d}{3^d} = 1 + 2^{2d+1}\,\frac{2^{M-1}-1}{3^d}, \tag{1}$$

an odd integer by Lemma 2.2 (Theorem 6.1 develops its structure; nothing beyond the formula and Lemma 2.6 below is needed before Section 6).

Definition 2.5. For $d \geq 1$ let $\beta_d(s)$ be the bit length of $T_s^{(d)} := w_d(1 + \mathrm{ord}_d\ s)$ and define the *level-d detector*

$$\lambda_d(x) = \max\{\, s \geq 1 :\ x \equiv T_s^{(d)} \pmod{2^{\,\beta_d(s)+1}} \,\} \ \in\ \mathbb{Z}_{\geq 1} \cup \{\bot\},$$

the maximum of the empty set being $\bot$.

Lemma 2.6. *Every template satisfies* $T_s^{(d)} \equiv 9 \pmod{32}$ *if* $d = 1$ *and* $T_s^{(d)} \equiv 1 \pmod{32}$ *if* $d \geq 2$. *Consequently* $\lambda_d(y) = \bot$ *for every* d *whenever* $y \not\equiv 1, 9 \pmod{32}$.

Proof. For $d \geq 2$, $2^{2d+1} \equiv 0 \pmod{32}$ in (1). For $d = 1$: M odd, $M \geq 3$, so $2^{M-1} \equiv 0 \pmod 4$, whence $(2^{M-1} - 1)/3 \equiv (-1) \cdot 3^{-1} \equiv 1 \pmod 4$ and $w_1(M) \equiv 1 + 8 \equiv 9 \pmod{32}$. Matching any template forces the mod-32 congruence, since every matching modulus $2^{\beta_d(s)+1}$ contains 32 ($\beta_d(s) \geq 4$). □

3 The master theorem: shadowing the -5 cycle

The map has the rational cycle $-5 \mapsto -7 \mapsto -5$: $f(-5) = -7$, $f(-7) = -5$, with multiplier $3^2/2^3 = \frac{9}{8} > 1$. Expanding cycles live at negative integers (the cycle equation gives $x = c/(2^E - 3^K)$ with $2^E < 3^K$), and their positive shadows gain value while walking the cycle's coordinate itinerary.

Lemma 3.1 (shadow period)**.** *Let* $N \geq 8$, w *odd, and* $x = 2^N w - 5$. *Then*

$$\mathrm{v}_2(3x + 1) = 1, \quad f(x) = 3 \cdot 2^{N-1} w - 7, \qquad \mathrm{v}_2(3f(x) + 1) = 2, \quad f^2(x) = 9 \cdot 2^{N-3} w - 5,$$

so that $8\, f^2(x) = 9x + 5$ *and* $f^2(x)/x = \frac{9}{8} + \frac{5}{8x}$. *Moreover* $\tau(x) = \tau(f^2(x)) = 2$ *and* $\tau(f(x)) = 1$*;* $x \equiv f^2(x) \equiv -5$ *and* $f(x) \equiv -7 \pmod{2^{N-3}}$*; and* $\lambda_d(x) = \lambda_d(f(x)) = \lambda_d(f^2(x)) = \bot$ *for every* d.

Proof. Direct computation: $3x+1 = 2(3 \cdot 2^{N-1} w - 7)$ with odd bracket; $3f(x)+1 = 4(9 \cdot 2^{N-3} w - 5)$ with odd bracket; $8f^2(x) = 9 \cdot 2^N w - 40 = 9x + 5$. Fuel: $x + 1 = 4(2^{N-2} w - 1)$; $f(x) + 1 = 2 \cdot 3(2^{N-2} w - 1)$ with odd second factor; $f^2(x) + 1 = 4(9 \cdot 2^{N-5} w - 1)$ (odd bracket, $N \geq 6$). Residues are immediate. Detectors: $x \equiv f^2(x) \equiv 27$ and $f(x) \equiv 25 \pmod{32}$ (using $N - 3 \geq 5$), and $27, 25 \notin \{1, 9\}$; apply Lemma 2.6. □

Theorem 3.2 (master no-go)**.** *Let* $m, d \geq 0$ *and let* g *be any real-valued function of* $\big(x \bmod 2^m,\ \tau(x),\ \mathrm{len}(x),\ \lambda_1(x), \ldots, \lambda_d(x)\big)$. *Then* $\Phi(x) = \log_2 x + g(\cdots)$ *is not nonincreasing along* f *on odd* $x > 1$.

Proof. Take $N \geq \max(m + 3, 8)$, $k \geq 3$, and $w = 2^{k-1} + 1$ in Lemma 3.1. Then additionally $\mathrm{len}(x) = \mathrm{len}(f^2(x)) = N + k$, since w places x in the lower part of its dyadic window: $9(2^k + 2) < 2^{k+4}$. Hence x and $f^2(x)$ have identical coordinate vectors while $f^2(x) > x$, and summing the two monotonicity constraints at x and $f(x)$ gives $0 \leq -\log_2\big(f^2(x)/x\big) < -\log_2 \frac{9}{8} < 0$. □

The smallest member ($N = 8$, $k = 3$) is $1275 \mapsto 1913 \mapsto 1435$, with $8 \cdot 1435 = 9 \cdot 1275 + 5$ and ratio product $\frac{287}{255}$: two integers refute every g on the full coordinate algebra at $m \leq 5$.

Remark 3.3 (general principle). Nothing is special about -5: every expanding rational cycle of the map — every solution of the cycle equation with $2^E < 3^K$, e.g. the $K = 7$ cycle at -17 with multiplier $3^7/2^{11}$ — shadows at positive integers $\equiv$ (cycle member) mod 2^N into a closed coordinate itinerary of value gain $3^K/2^E > 1$, refuting every correction that is 2-adically local along it. Expanding negative cycles are certificate factories against local potentials.

4 Iterating the shadow: leading digits

A coordinate can defeat a single shadow period only by separating x from $f^2(x) = \frac{9}{8}x + \frac{5}{8}$; the leading digits do (the ratio $\frac{9}{8}$ shifts them). Iteration removes this refuge.

Theorem 4.1 (leading digits do not survive)**.** *For every $m, j, d \geq 0$ and every g, the potential $\log_2 x + g\big(x \bmod 2^m,\ \tau(x),\ \mathrm{LD}_j(x),\ \lambda_1(x), \ldots, \lambda_d(x)\big)$ is not nonincreasing along f on odd $x > 1$.*

Proof. For $x \equiv -5 \pmod{2^{3a+c}}$, $c \geq 5$, Lemma 3.1 applies a times (each period consumes three bits of shadow depth), the itinerary alternating residues $-5, -7$ with fuel $2, 1$ and every detector $\perp$ throughout, giving the exact closed form

$$f^{2a}(x) = \frac{9^a x + 5\,(9^a - 8^a)}{8^a} = \left(\tfrac{9}{8}\right)^a x\,(1+\delta), \qquad 0 < \delta < \tfrac{5}{x}.$$

Now $\beta = \log_2 \frac{9}{8}$ is irrational ($2^{p+3q} = 3^{2q}$ is impossible: v_3 of the sides differ), so by Dirichlet there is $a \geq 1$ with $\|a\beta\| < 2^{-j-3}$ (distance to the nearest integer P; note $a\beta > 0$ regardless of the side). Choose $N = 3a + \max(m, 5) + j + 8$ and a free odd multiplier w placing the fractional part of $\log_2 x$, $x = 2^N w - 5$, at distance $> 2^{-j-2}$ from every breakpoint of LD_j (possible since $\log_2(2^N w - 5) \approx N + \log_2 w$ and $\{\log_2 w\}$ is dense), with N large enough that $\log_2(1+\delta) < 2^{-j-3}$. Then $\log_2 f^{2a}(x) = \log_2 x + P \pm \|a\beta\| + \log_2(1+\delta)$, so LD_j closes; all other coordinates close by the itinerary; and telescoping monotonicity over the $2a$-step walk gives $0 \geq a\beta + \log_2(1+\delta) > 0$. □

A concrete instance ($j = 4$): $a = 53$ gives $53\beta = 9.0060\ldots$; the verifier realizes the full 106-step walk in exact integers at depth $N = 177$, with top-4 bits `1011` → `1011`, all coordinates frozen, and value gain $2^{9.006}$.

5 Bounded corrections

Theorem 5.1 (no bounded correction)**.** *Let $G : 2\mathbb{Z} + 1 \to \mathbb{R}$ be any bounded function. Then $\log_2 x + G(x)$ is not nonincreasing along f on odd $x > 1$.*

Proof. Let $\Omega = \sup G - \inf G$. Telescoping monotonicity over the burn of $2^M - 1$ (Lemma 2.3 iterated: $M-1$ steps, each of ratio $> \frac{3}{2}$) forces $(M-1)\log_2 \frac{3}{2} \leq \Omega$, false for $M > 1 + \Omega/\log_2 \frac{3}{2}$. □

Theorems A–C partition the correction landscape: bounded corrections fail on the burn alone; unbounded corrections through residues, fuel, length, leading digits, and suffix detectors of every depth fail on the shadow. Section 7 treats the size of x read to an arbitrary fixed precision, and Section 11 describes what is left.

6 The Mersenne ancestry tower

We now develop the structure behind the detectors. Define $w_0(M) = 2^M - 1$ and $w_d(M)$ as in (1).

Theorem 6.1. *For $d \geq 1$ and $M \equiv 1 \pmod{\mathrm{ord}_d}$, $M > 1$:*

(a) (Domain.) *3^d divides $2^{M+2d} - 2^{2d+1} + 3^d$ iff $M \equiv 1 \pmod{\mathrm{ord}_d}$.*

(b) (Form.) *$w_d(M)$ is an odd integer > 1 with $\tau(w_d(M)) = 1$.*

(c) (Descent.) $3\,w_d(M)+1 = 4\,w_{d-1}(M)$*; hence* $\mathrm{v}_2(3w_d(M)+1)=2$*,* $f(w_d(M)) = w_{d-1}(M)$*, and* $f^d(w_d(M)) = 2^M-1$*, each step of value ratio* $\frac{3}{4}+\frac{1}{4w_d}\in(\frac{3}{4},1)$*.*

(d) (Normal form.) $B_d := (2^{\mathrm{ord}_d}-1)/3^d$ *is an integer in* $(0, 2^{\mathrm{ord}_d})$ *by Lemma 2.1, and for* $M = 1+\mathrm{ord}_d\ s$*,*

$$w_d(M) = 1 + 2^{2d+1}\cdot\big[\textit{the } \mathrm{ord}_d\textit{-bit block of } B_d\big]^s,$$

the bracket denoting s *carry-free repetitions: an exactly periodic binary word of period* $2\cdot 3^{d-1}$*.*

(e) (Freeze and separation.) $w_d(M)\equiv\Lambda_d := 1-2^{2d+1}3^{-d} \pmod{2^{M-1}}$*; and for* $d\neq d'$*,* $\mathrm{v}_2(\Lambda_d-\Lambda_{d'}) = 2\min(d,d')+1$*.*

Proof. (a) Modulo 3^d the numerator is $2^{2d}(2^M-2)$, so divisibility is $2^{M-1}\equiv 1 \pmod{3^d}$; apply Lemma 2.2. (b) The second form in (1) is 1 plus an even number, and $w_d+1 = 2(1+2^{2d}q)$ with $2^{2d}q$ even. (c) Clearing 3^{d-1}, the identity reduces to $3^d+3^{d-1} = 4\cdot 3^{d-1}$; w_{d-1} is odd, so $\mathrm{v}_2=2$ exactly; iterate. (d) With $M-1=\mathrm{ord}_d\ s$, $(2^{M-1}-1)/3^d = B_d\sum_{i=0}^{s-1}2^{\mathrm{ord}_d\, i}$, a base-$2^{\mathrm{ord}_d}$ number all of whose digits equal $B_d<2^{\mathrm{ord}_d}$: no carries; the $+1$ and the shift by 2^{2d+1} cannot interact. (e) $3^d\cdot\frac{2^{M-1}-1}{3^d}\equiv -1 \pmod{2^{M-1}}$; and for $d>d'$, $\Lambda_d-\Lambda_{d'} = 2^{2d'+1}3^{-d}\big(3^{\,d-d'}-2^{\,2(d-d')}\big)$ with odd bracket. □

Levels $0,1,2$ are the Mersennes, the family $x'_M = (2^{M+2}-5)/3 = (10)^{(M-3)/2}1001$ with $f(x'_M)=2^M-1$, and $(111000)^k01$; the tower is the complete uniform-e=2 ancestry of maximal fuel, and every tower member lies on the residue class $8y+1$. Composing the tower with the burn (Remark 2.4) yields, for each (d,M), an exactly known orbit segment of length $d+M+1$ terminating on the base-3 repunit $(3^M-1)/2$.

Lemma 6.2. *(1)* $\lambda_d(T_s^{(d)}) = s$*: the detector reads its own tower index, unboundedly. (2) For* $d\neq d'$*, with* $\nu = 2\min(d,d')+1$*: for all* $M>\nu+1$ *in the level-d progression,* $\lambda_{d'}(w_d(M))\le \nu/\mathrm{ord}_{d'}$*.*

Proof. (1) At index s the congruence is an equality that holds; at $s'>s$ the modulus exceeds both operands and templates strictly increase. (2) Suppose $w_d(M)$ matches $T_s^{(d')}$ with $M' = 1+\mathrm{ord}_{d'}\ s$ and, toward a contradiction, $M'-1\ge\nu+1$. The matching modulus refines $2^{\nu+1}$ (as $\beta_{d'}(s)\ge M'$), so Theorem 6.1(e) on both sides forces $\Lambda_d\equiv\Lambda_{d'} \pmod{2^{\nu+1}}$, contradicting $\mathrm{v}_2(\Lambda_d-\Lambda_{d'})=\nu$. Hence $\mathrm{ord}_{d'}\ s\le\nu$. (Small-index templates have not converged to $\Lambda_{d'}$; this affects which small values occur, not the bound.) □

Remark 6.3 (the family method and the price list). Historically, Theorems A–C were first obtained for subclasses by a family method that yields quantitative structure the shadow does not. The *burn* family 3^j2^t-1 ascends at ratio $>\frac{3}{2}$ per step with frozen residue -1 and $\lambda\equiv\perp$, forcing g to grow at slope $\log_2\frac{3}{2}$ along a coordinate line; the tower's e=2 chains cap the same values from above, each level costing one $\log_2\frac{4}{3}$. The resulting price list — each detector level buys a would-be potential exactly $\log_2\frac{4}{3}$ of budget against an unbounded liability — explains *why* suffix information fails, while the shadow certificates prove *that* it fails. The full family proofs are retained in the repository.

7 The quantized logarithm

Theorems A and B leave one natural coordinate untouched: the size of x read to a fixed precision, integer and fractional parts of $\log_2 x$ jointly. Write $Q = 2^j$ and

$$c(x) = \lfloor Q \log_2 x \rfloor,$$

the *quantized logarithm* at precision j ($c(x) = \mathrm{len}(x) - 1$ for $j = 0$). A single shadow chain cannot close it: one period of the shadow gains $\log_2 \frac{9}{8} > 2^{-j}$ for $j \geq 3$, so x and $f^{2a}(x)$ never share a cell. The remedy is to use several witnesses, each a fresh integer: at every junction the position inside the cell is reset, and the total displacement around the cycle can be made zero exactly. This section proves that such certificates exist at every precision, for arbitrary moduli and detectors at once, and that their minimal length is a Diophantine invariant of $\log_2 3$.

Throughout, $\beta = \log_2 3$, $L = \lfloor Q\beta \rfloor$, $\varphi = Q\beta - L \in (0,1)$ (irrational β), and for odd x

$$\theta(x) = Q \log_2 x - c(x) \in [0,1), \qquad \eta(x) = Q \log_2\Big(1 + \frac{1}{3x}\Big) \in \Big(0, \frac{Q}{3x \ln 2}\Big).$$

Definition 7.1. Let C be any function on the odd integers (a *coordinate*). A *closed certificate* of length K for C is a list of odd integers $x_1, \ldots, x_K > 1$, not necessarily distinct, with

$$C(f(x_i)) = C(x_{i+1}) \quad (i \bmod K) \qquad \text{and} \qquad \prod_{i=1}^{K} f(x_i) > \prod_{i=1}^{K} x_i.$$

Lemma 7.2. *If a closed certificate exists for C, then no potential $\log_2 x + g(C(x))$ is nonincreasing along f on odd $x > 1$.*

Proof. Sum the K inequalities $\Phi(f(x_i)) \leq \Phi(x_i)$. The g-terms cancel in pairs by closure, leaving $\sum_i \log_2\big(f(x_i)/x_i\big) \leq 0$, which contradicts the gain. □

The number of witnesses is governed by the following quantity.

Definition 7.3. $K \geq 1$ is *admissible at precision Q* if

$$s(Q,K) := Q\lfloor K\beta \rfloor - KL \;\geq\; 0, \qquad \text{equivalently} \qquad \frac{\lfloor K\beta \rfloor}{K} \;\geq\; \frac{\lfloor Q\beta \rfloor}{Q}.$$

The identity $s(Q,K) = K\varphi - Q\{K\beta\}$ shows $s(Q,K) < K$ always, and $s(Q,Q) = 0$, so $K = Q$ is admissible at every precision.

Two lemmas

Lemma 7.4 (displacement)**.** *Let x be odd with $e = \mathrm{v}_2(3x+1)$. Then*

$$c(f(x)) = c(x) + L - Qe + \delta, \qquad \delta = \big\lfloor \theta(x) + \varphi + \eta(x) \big\rfloor.$$

In particular, if $\eta(x) \leq 1 - \varphi$ then $\delta \in \{0,1\}$, with $\delta = 0$ whenever $\theta(x) < 1 - \varphi - \eta(x)$ and $\delta = 1$ whenever $\theta(x) \geq 1 - \varphi$. The last condition is the exact integer inequality $(3x)^Q \geq 2^{\,c(x)+1+L}$.

Proof. $Q \log_2 f(x) = Q \log_2 x + Q\beta - Qe + \eta(x) = c(x) + L - Qe + \big(\theta(x) + \varphi + \eta(x)\big)$. For the integer form: $(3x)^Q \geq 2^{c+1+L}$ iff $Q\beta + c + \theta \geq c + 1 + L$ iff $\theta \geq 1 - \varphi$. □

Lemma 7.5 (rational cycles)**.** *Let $(e_1, \dots, e_K)$ be any word of positive integers with sum E, and put $E_i = e_1 + \cdots + e_i$. Then*

$$y_1 = \frac{\sum_{i=1}^{K} 3^{K-i}\, 2^{E_{i-1}}}{2^E - 3^K}$$

is a rational number whose denominator is prime to 6; its orbit under f (extended to $\mathbb{Z}_2$) is purely periodic, $y_{i+1} = (3y_i + 1)/2^{e_i}$ and $y_{K+1} = y_1$, with $\mathrm{v}_2(3y_i + 1) = e_i$ exactly for every i. Every y_i is a 2-adic unit and a 3-adic integer.

Proof. Composing the affine maps $y \mapsto (3y + 1)/2^{e_i}$ gives $2^E y_{K+1} = 3^K y_1 + \sum_i 3^{K-i} 2^{E_{i-1}}$, so y_1 as stated is the unique fixed point of the composite; the same formula applied to the rotated words shows that every y_i has the shape $c_i/(2^E - 3^K)$ with $c_i \in \mathbb{Z}$, so every $y_i \in \mathbb{Z}_2 \cap \mathbb{Z}_3$. If some y_{i+1} were even, then $3y_{i+1} + 1$ would be a 2-adic unit and $y_{i+2} = (3y_{i+1} + 1)/2^{e_{i+1}} \notin \mathbb{Z}_2$, a contradiction. Hence every y_{i+1} is odd, i.e. $\mathrm{v}_2(3y_i + 1) = e_i$. (Periodic parity sequences of rational 2-adic integers are classical; see Lagarias [6], Theorem L.) □

For the word $(1, 2)$ the cycle is $-5 \mapsto -7$; for (2) it is the fixed point 1. Lemma 3.1 is the shadow of $(1, 2)$; the proof below shadows the cycle of a word chosen to fit the precision.

The theorem

Theorem 7.6 (the quantized logarithm does not survive)**.** *Let $j, m, d \geq 0$ and $N \geq 1$, and let g be any real-valued function of*

$$\big(\lfloor 2^j \log_2 x \rfloor,\ \tau(x),\ x \bmod N,\ \lambda_1(x), \dots, \lambda_d(x)\big).$$

Then $\log_2 x + g(\cdots)$ is not nonincreasing along f on odd $x > 1$. More precisely, write $N = 2^m 3^u N'$ with $\gcd(N', 6) = 1$. For every $K \geq 2$ admissible at $Q = 2^j$ with $2\lfloor K\beta \rfloor \geq 3K$ — a condition satisfied by every admissible $K \geq 2$ when $j \geq 1$ — there is a closed certificate for the coordinate of length tK, where $t \geq 1$ depends only on K and N' and $t = 1$ when $N' = 1$. Its witnesses can be taken of bit length at most

$$\Theta = m + \log_2(3^u N') + 6tK + j + \log_2 \frac{1}{\min(\varphi, 1 - \varphi)} + 10.$$

Proof. Put $E = \lfloor K\beta \rfloor$ and $s = s(Q, K) \in [0, K)$.

Step 1: the word. Let $p = 5K - 3E$ and $r = 2E - 3K$. Both are nonnegative ($\beta < \frac{5}{3}$ and the hypothesis; for $j \geq 1$ the hypothesis follows from admissibility, since $L/Q \geq \frac{3}{2}$ for $Q = 2, 4, 8$ and $Q\beta - 1 \geq \frac{3}{2}Q$ for $Q \geq 12$), and the cyclic word

$$(e_1, \dots, e_K) = (2, 1)^p\, (3, 1, 1)^r$$

has length $2p + 3r = K$ and sum $3p + 5r = E$. It contains neither the factor $(2, 2)$ nor the factor $(2, 1, 1)$, read cyclically. Let $y_1, \dots, y_K$ be its rational cycle (Lemma 7.5).

Step 2: detectors and fuel on the cycle. The parity word of y_i under T (Section 10) begins $\mathtt{O}\,\mathtt{E}^{e_i - 1}\,\mathtt{O}\,\mathtt{E}^{e_{i+1} - 1} \cdots$, and $y_i \bmod 32$ is determined by its first five letters (Terras [13]: the first p letters and the residue modulo 2^p determine each other). The residues 1 and 9 have parity words `OEOEO` and `OEOOO`, which would force $(e_i, e_{i+1}) = (2, 2)$, resp. $(e_i, e_{i+1}, e_{i+2}) = (2, 1, 1)$. Neither occurs, so no y_i is $\equiv 1$ or 9 (mod 32), and by Lemma 2.6 every λ_d is $\perp$ on every odd integer $\equiv y_i \pmod{32}$. Lemma 2.3 holds verbatim for 2-adic units: a member with $e_i \geq 2$ has

$\tau = 1$, and along a run of $e = 1$ steps τ decreases by one per step. The runs of 1s in the word have length at most 2, so $\tau(y_i) \le 3$ for all i.

Step 3: residues. Put $M = \max(m, 5) + E$. On $\mathbb{Z}/N'$ the maps $\rho \mapsto (3\rho + 1)2^{-e_i}$ are bijections; let A be their composite over one period and t its order as a permutation. Fix $\rho_1 \in \mathbb{Z}/N'$ and define $\rho_{i+1} = (3\rho_i + 1)2^{-e_i}$ for $i = 1, \dots, tK$ (indices of e taken modulo K), so $\rho_{tK+1} = \rho_1$. For $1 \le i \le tK$ let R_i be the residue class modulo $R = 2^M 3^u N'$ of integers x with

$$x \equiv y_i \pmod{2^M}, \qquad x \equiv y_i \pmod{3^u}, \qquad x \equiv \rho_i \pmod{N'}$$

(indices of y modulo K; the first two congruences make sense because $y_i \in \mathbb{Z}_2 \cap \mathbb{Z}_3$). Every element of R_i is odd. For $x \in R_i$: $3x + 1 \equiv 3y_i + 1 \pmod{2^M}$, so $\mathrm{v}_2(3x + 1) = e_i$ and $f(x) \equiv y_{i+1} \pmod{2^{M-e_i}}$ with $M - e_i \ge \max(m, 5)$; $3x+1 \equiv 3y_i+1 \pmod{3^{u+1}}$, so $f(x) \equiv y_{i+1} \pmod{3^u}$; and $f(x) \equiv \rho_{i+1} \pmod{N'}$. Hence for any $x \in R_i$ and $x' \in R_{i+1}$: $f(x) \equiv x' \pmod{N}$; $\tau(f(x)) = \tau(y_{i+1}) = \tau(x')$ since $\tau(y_{i+1}) \le 3 < M - e_i$; and all detectors are $\perp$ at both $f(x)$ and x' by Step 2. Every coordinate other than c closes between R_i and R_{i+1}.

Step 4: cells. Set $\delta_i = 1$ if $i \bmod K \in \{1, \dots, s\}$ and $\delta_i = 0$ otherwise, so that each period carries exactly s times. Choose $c_1 = H$ and $c_{i+1} = c_i + L - Qe_i + \delta_i$. Then $c_{tK+1} = c_1 + t(KL - QE + s) = c_1$. Since $e_i \le 3$ and $L \ge Q$, the downward excursion satisfies $c_1 - c_i \le t \sum_{i \le K} \max(0, Qe_i - L) \le 2tKQ$.

Step 5: witnesses. Let $\ell_0 = \min(\varphi, 1 - \varphi)$ and take $H \ge Q\big(\log_2 R + j + \log_2 \frac{2}{\ell_0 \ln 2} + 2tK + 1\big)$. For each i, the integers in cell c_i with $\theta < 1 - \varphi - \eta^*$ (if $\delta_i = 0$), resp. $\theta \ge 1 - \varphi$ (if $\delta_i = 1$), where $\eta^* = Q/(3 \cdot 2^{c_i/Q} \ln 2) \ge \sup_{\mathrm{cell}} \eta$, form an interval of length at least $2^{c_i/Q}\, \ell \ln 2/Q$ with $\ell \ge \ell_0/2$ (the choice of H gives $\eta^* \le (1 - \varphi)/2$), which is at least R by the choice of H and Step 4; so the interval contains an element x_i of R_i. By Lemma 7.4, $c(f(x_i)) = c_i + L - Qe_i + \delta_i = c_{i+1} = c(x_{i+1})$, and with Step 3 the coordinate closes at every junction. The upward excursion is at most $t \sum_{i \le K} \max(0, L - Qe_i + 1) \le tK(L - Q + 1)$, and with $M \le m + 5 + E$, $E < \frac{8}{5}K$ and $L - Q < \frac{3}{5}Q$ the bit length of every witness is at most Θ when H is taken minimal.

Step 6: gain. With $y_i' = f(x_i)$,

$$\frac{\prod y_i'}{\prod x_i} = \frac{3^{tK}}{2^{tE}} \prod_i \Big(1 + \frac{1}{3x_i}\Big) > \Big(\frac{3^K}{2^E}\Big)^t > 1,$$

since $E = \lfloor K\beta \rfloor < K\beta$. Lemma 7.2 finishes the proof. □

Remark 7.7. (i) Nothing in the proof uses that the witnesses are distinct, and no estimate of the correction terms η beyond $\eta \le (1 - \varphi)/2$ is needed: the gain is automatic from $2^E < 3^K$. (ii) Without detectors ($d = 0$) any word over $\{1, 2\}$ with sum E serves, and then every admissible $K \ge 2$ is realised, also for $j = 0$; M must then be raised to $\max(m, 5, 2K - E + 2) + E$ to cover the longer fuel runs. (iii) At $j = 0$ and $K = 2$ the word is $(2, 1)$, the cycle is $-7 \mapsto -5$, and the construction reproduces the shadow of Lemma 3.1 with its two steps now read as two witnesses. Theorem 3.2 is the case $j = 0$, $N = 2^m$ of Theorem 7.6. (iv) The smallest instances are tiny. At $(j, m) = (3, 4)$, $K = 2$: $4123 \mapsto 6185$, $5817 \mapsto 4363$ close with $c(6185) = c(5817) = 100$, $c(4363) = c(4123) = 96$, and $6185 \cdot 4363 > 4123 \cdot 5817$. The pairs $x_1 = 2^n + 27$, $n \ge 10$, with x_2 the least integer $\equiv 9 \pmod{16}$ in the non-carrying part of $c(f(x_1))$ whose image returns to $c(x_1)$ with matching τ, give certificates at every height. The cost law below says why $K = 2$ suffices there and why $K = 7$ is needed at $j = 5$.

The cost of precision

Theorem 7.6 uses admissibility to construct certificates. For witnesses that are not too small it is also necessary.

Proposition 7.8 (closure criterion)**.** *Let $x_1, \dots, x_K$ be a closed certificate for $\big(c(x), \tau(x), x \bmod 2^m\big)$ with valuations $e_i = \mathrm{v}_2(3x_i + 1)$ and $E = \sum e_i$, and suppose every $x_i \geq \max\big(Q/(3(1-\varphi)\ln 2),\ K/(3(1-\{K\beta\})\ln 2)\big)$. Then $QE - KL \in [0, K]$, $E \leq \lfloor K\beta \rfloor$, and K is admissible at Q. Consequently, for $j \geq 1$ and $K \geq 2$, closed certificates of length K with all witnesses above any prescribed height exist if and only if K is admissible.*

Proof. The first height condition gives $\eta(x_i) \leq 1 - \varphi$, so Lemma 7.4 applies with $\delta_i \in \{0, 1\}$; summing the displacements around the cycle, $KL - QE + \sum \delta_i = 0$ with $\sum \delta_i \in [0, K]$. The gain is $2^{-E}3^K \prod(1 + 1/(3x_i)) < 2^{K\beta - E} \exp\big(K/(3 \min x_i)\big)$, so gain > 1 forces $E - K\beta < K/(3 \ln 2 \min x_i) \leq 1 - \{K\beta\}$ by the second height condition, i.e. $E \leq \lfloor K\beta \rfloor$. Then $s(Q, K) = Q\lfloor K\beta \rfloor - KL \geq QE - KL \geq 0$. The converse is Theorem 7.6. □

Proposition 7.9 (cost law)**.** *Let $K_{\min}(Q)$ be the least admissible K at precision Q. Then $K_{\min}(Q) \leq Q$, and $\lfloor K_{\min}\beta \rfloor / K_{\min}$ is a best approximation to β from below among fractions with denominator at most $K_{\min}$; hence $K_{\min}(Q)$ is the denominator of a convergent or intermediate fraction of the continued fraction of $\log_2 3$ lying below $\log_2 3$. For $Q = 2^j$, $j = 0, 1, \dots, 10$,*

$$K_{\min} = 1,\ 2,\ 2,\ 2,\ 7,\ 7,\ 12,\ 12,\ 12,\ 53,\ 665.$$

Proof. $K = Q$ is admissible. If $K^* = K_{\min}(Q)$ then for $K < K^*$, $\lfloor K\beta \rfloor / K < L/Q \leq \lfloor K^*\beta \rfloor / K^*$; as $\lfloor K\beta \rfloor / K$ is the largest fraction with denominator K below β, $\lfloor K^*\beta \rfloor / K^*$ is the largest fraction with denominator at most K^* below β. One-sided best approximations are exactly the convergents and intermediate fractions on that side (Khinchin [4], Ch. I). The table is an exact integer computation, $\lfloor n\beta \rfloor = \mathrm{len}(3^n) - 1$. □

Thus the price of precision is not a continuous quantity but a Diophantine one: it jumps at the semiconvergent denominators $2, 7, 12, 53, 665, \dots$ of $\log_2 3$, and between jumps extra bits of precision are free. The certificate miner of Section 9, which knows nothing of this, finds cycles of length exactly $K_{\min}$ on the quantized-log windows it is given (2 at $j = 3$, 7 at $j = 5$).

Arbitrary moduli and multi-step potentials

Theorem 7.6 already allows an arbitrary modulus. The single-orbit theorems extend the same way.

Corollary 7.10 (arbitrary modulus)**.** *Theorem 4.1 remains true with $x \bmod 2^m$ replaced by $x \bmod N$ for any $N \geq 1$. The class of Theorem 3.2 with $x \bmod N$ is contained in the class of Theorem 7.6 at $j = 0$.*

Proof. Write $N = 2^m 3^u N'$ with $\gcd(N', 6) = 1$. On the shadow $x = 2^{N_0} w - 5$ of Theorem 4.1, $f^{2a}(x) = \big(9^a x + 5(9^a - 8^a)\big)/8^a$. Let o be the multiplicative order of $9 \cdot 8^{-1}$ modulo N'; if $o \mid a$ then $9^a \equiv 8^a \pmod{N'}$ and $f^{2a}(x) \equiv x \pmod{N'}$ for every shadow x. If $2a \geq u$ then $9^a \equiv 0 \pmod{3^u}$ and $f^{2a}(x) \equiv -5 \pmod{3^u}$, so it suffices to take $3^u \mid w$. In the proof of Theorem 4.1 apply Dirichlet to $o\beta$ instead of β to obtain $a = oa'$ with $\|a\beta\| < 2^{-j-3}$, and choose the free multiplier w among odd multiples of 3^u (still dense in the logarithmic scale). The second statement is Remark 7.7(iii). □

Corollary 7.11 (multi-step potentials)**.** *Fix $k \geq 1$. Theorems 4.1 and 7.6, and hence the class of Theorem 3.2, remain true for potentials that are merely nonincreasing along f^k, i.e. with $\Phi(f^k(x)) \leq \Phi(x)$ for all odd $x > 1$ in place of $\Phi(f(x)) \leq \Phi(x)$.*

Proof. For Theorem 4.1, apply Dirichlet to $k\beta$ to get $a = ka'$ with $\|a\beta\| < 2^{-j-3}$; the $2a$-step walk of the proof then decomposes into $2a'$ segments of k steps, and summing the k-step inequalities along them gives the same contradiction. For Theorem 7.6, run the proof with a word of length $K = kK_0$ and $E = \lfloor K\beta \rfloor$, where $K_0 = Q$ if $j \geq 1$ and $K_0 = 2$ if $j = 0$ (then $2E \geq 3K$ in both cases), and take as witnesses only the segment starts $x_i \in R_{(i-1)k+1}$, $i = 1, \ldots, tK_0$, so that the edges are $x_i \mapsto f^k(x_i)$. Over a segment with valuation sum E_i, Lemma 7.4 iterated gives $c(f^k(x)) = c(x) + \lfloor kQ\beta \rfloor - QE_i + \delta$ with $\delta \in \{0, 1\}$ decided by $\theta(x)$ against $1 - \{kQ\beta\}$, and closure needs $\sum \delta = QE - K_0 \lfloor kQ\beta \rfloor$, which lies in $[0, K_0]$ for the stated K_0 (it vanishes for $K_0 = Q$). The rest of the proof is unchanged, with $\{kQ\beta\}$ in place of φ throughout. □

What is *not* covered is noted in Section 11: the leading-digit coordinate LD_j taken jointly with the quantized logarithm, and corrections that depend on the orbit rather than on x.

8 How often must a local potential fail?

Theorems A–E exhibit failures of every local potential. We now ask how rare the failures can be made, and find that the answer is a sharp exponential rate: a depth-m local correction can fail on a set of density about 2^{-cm} with $c = 0.0500\ldots$, and no depth-m local correction can do better by more than a factor polynomial in m. The lower bound is the shadow mechanism applied to *all* expanding cycles of a given period at once; the upper bound is a potential that simply looks ahead along the orbit as far as m bits allow. The constant c is the large-deviation exponent for the event that the orbit has not decreased after m parity steps.

Statement

For $x \in \mathbb{Z}_2$ let $p_0(x)p_1(x)\cdots$ be the parity word of Section 10, $p_i(x) = T^i(x) \bmod 2$ with letters $\mathtt{O} = 1$, $\mathtt{E} = 0$.

Lemma 8.1 (Terras [13])**.** *For $n \geq 1$ the map $x \bmod 2^n \mapsto (p_0(x), \ldots, p_{n-1}(x))$ is a bijection from $\mathbb{Z}/2^n\mathbb{Z}$ onto $\{\mathtt{O}, \mathtt{E}\}^n$. In particular, for odd x the letters $p_1, \ldots, p_{n-1}$ are independent fair coin tosses when x is uniform on the odd residues modulo 2^n.*

A *depth-m local potential* is $\Phi_g(x) = \log_2 x + g(x \bmod 2^m)$ with g any real function on the odd residues. Its *failure set* is

$$F(g) = \{\, x \text{ odd},\ x > 1 :\ \Phi_g(f(x)) > \Phi_g(x) \,\},$$

and densities are taken among the odd integers: $\underline{d}(A) = \liminf_X \#(A \cap [1, X])/(X/2)$. Put

$$\theta = \frac{1}{\log_2 3} = 0.6309\ldots, \qquad H(p) = -p \log_2 p - (1-p)\log_2(1-p),$$

$$c = 1 - H(\theta) = 0.05004\ldots.$$

Theorem 8.2 (failure density)**.** *Let $m \geq 12$.*

1. *For every g, the failure set $F(g)$ contains a union of residue classes of odd integers whose density is at least $2^{-cm}/\big(16(m+2)^5\big)$.*

2. *There is a g for which $F(g)$ is contained in the union of a set of natural density at most $4\cdot 2^{-cm}$ and a finite set.*

Consequently $\delta_m := \inf_g \underline{d}(F(g))$ satisfies

$$\frac{2^{-cm}}{16(m+2)^5} \;\le\; \delta_m \;\le\; 4\cdot 2^{-cm}, \qquad \text{and} \qquad \lim_{m\to\infty}\frac{1}{m}\log_2\delta_m = -c.$$

The lower bound

Proof of Theorem 8.2(1). Put $P = m+1$ and fix g. For a word $w \in \{\mathtt{O},\mathtt{E}\}^P$ let $k(w)$ be its number of $\mathtt{O}$s; call w *expanding* if $3^{k(w)} > 2^P$, i.e. $k(w) > \theta P$. Let $\rho = \lceil 3\log_2 P\rceil$ and let $\mathcal{C}$ be the set of rotation classes $[w]$ of expanding words all of whose cyclic runs of $\mathtt{E}$ have length at most ρ.

Step 1: cycles. For $[w]\in\mathcal{C}$ pick the representative w beginning with $\mathtt{O}$ (it has one, being expanding) and read it as $\mathtt{O}\mathtt{E}^{e_1-1}\mathtt{O}\mathtt{E}^{e_2-1}\cdots\mathtt{O}\mathtt{E}^{e_K-1}$ with $K = k(w)$ and $e_1+\cdots+e_K = P$. Lemma 7.5 gives the rational cycle $y_1,\dots,y_K$ with $\mathrm{v}_2(3y_i+1) = e_i$; the parity word of y_1 is w^∞.

Step 2: edge classes. Let A_i be the set of odd $x>1$ with $x \equiv y_i \pmod{2^{m+e_i}}$. For $x\in A_i$ we have $\mathrm{v}_2(3x+1) = e_i$ and $f(x)\equiv y_{i+1} \pmod{2^m}$, so the monotonicity constraint at x reads

$$g(y_{i+1} \bmod 2^m) - g(y_i \bmod 2^m) \;\le\; -(\log_2 3 - e_i) - \varepsilon(x), \qquad \varepsilon(x) = \log_2\Big(1+\frac{1}{3x}\Big) > 0.$$

Step 3: one class fails entirely. If no A_i were contained in $F(g)$, each constraint would hold at some $x_i\in A_i$, and summing over i the left sides telescope to 0 while the right sides sum to less than $-(K\log_2 3 - P) < 0$, a contradiction. So some $A_{i(w)} \subseteq F(g)$. Its density among odd integers is $2^{-(m+e_{i(w)}-1)} \ge 2^{-(m+\rho)}$, because every $e_i - 1$ is a cyclic run of $\mathtt{E}$ in w.

Step 4: disjointness. By Lemma 8.1, $A_{i(w)}$ is the set of odd integers whose parity word begins with a specific window of w^∞ of length $m+e_{i(w)} \ge P$. Such a window contains P consecutive letters of w^∞, a rotation of w. Distinct classes in $\mathcal{C}$ have no rotation in common, so the sets $A_{i(w)}$, $[w]\in\mathcal{C}$, are pairwise disjoint, and

$$\underline{d}(F(g)) \;\ge\; |\mathcal{C}|\cdot 2^{-(m+\rho)}.$$

Step 5: counting. Let $k_0 = \lceil \theta P\rceil$ (θP is not an integer). The number W of expanding words satisfies $W \ge \binom{P}{k_0} \ge 2^{PH(k_0/P)}/(P+1)$, and since H is decreasing on $[\frac12,1]$ with $|H'|\le 2$ on $[\theta,\theta+\frac{1}{13}]$, $H(k_0/P) \ge H(\theta) - 2/P$, whence $W \ge 2^{PH(\theta)}/(4(P+1))$. An expanding word with a cyclic run of $\rho+1$ letters $\mathtt{E}$ is determined by the position of that run and the remaining $P-\rho-1$ letters, which contain at least k_0 letters $\mathtt{O}$; by the entropy bound $\sum_{k\ge\lambda n}\binom{n}{k} \le 2^{nH(\lambda)}$ ($\lambda \ge \frac12$) and monotonicity of H, there are at most $P\cdot 2^{(P-\rho-1)H(\theta)} \le 2^{PH(\theta)}P^{1-3H(\theta)}$ of them, and $1-3H(\theta) < -1.849$. For $P\ge 13$ one has $P^{1.849} \ge 8(P+1)$, so at least $2^{PH(\theta)}/(8(P+1))$ expanding words have no long run, and $|\mathcal{C}| \ge 2^{PH(\theta)}/(8P(P+1))$. With $2^{-\rho}\ge 1/(2P^3)$ and $PH(\theta) - m = -cm + H(\theta) > -cm$,

$$\underline{d}(F(g)) \;\ge\; \frac{2^{PH(\theta)-m}}{16P^4(P+1)} \;\ge\; \frac{2^{-cm}}{16(m+2)^5}. \qquad \square$$

The upper bound: looking ahead

Let $\gamma(\mathtt{O}) = \log_2 3 - 1$ and $\gamma(\mathtt{E}) = -1$, the logarithmic gain of a T-step up to the term ε, and for any residue r modulo 2^m with parity word $p_0 \cdots p_{m-1}$ set

$$h(r) = \sum_{l=0}^{m-1} \Big(1 - \frac{l+1}{m}\Big)\gamma(p_l), \qquad \Psi(x) = \log_2 x + h(x \bmod 2^m) \quad (x \geq 1).$$

Lemma 8.3 (lookahead identity)**.** *For every integer $x \geq 1$ with parity word $p_0 p_1 \cdots$ and $k = \#\{l < m : p_l = \mathtt{O}\}$,*

$$\Psi(T(x)) - \Psi(x) = \frac{k \log_2 3 - m}{m} + [x \text{ odd}]\,\varepsilon(x).$$

Proof. The parity word of $T(x)$ is $p_1 p_2 \cdots$, and $\log_2(T(x)/x) = \gamma(p_0) + [x \text{ odd}]\varepsilon(x)$. Hence $\Psi(T(x)) - \Psi(x) - [x \text{ odd}]\varepsilon(x) = \gamma(p_0) + \sum_{l=1}^{m}\big(1 - \frac{l}{m}\big)\gamma(p_l) - \sum_{l=0}^{m-1}\big(1 - \frac{l+1}{m}\big)\gamma(p_l) = \frac{1}{m}\sum_{l=0}^{m-1}\gamma(p_l)$, the coefficient of $\gamma(p_m)$ being $1 - \frac{m}{m} = 0$. Finally $\sum_{l<m}\gamma(p_l) = k\log_2 3 - m$. □

Proof of Theorem 8.2(2). Take $g = h$ on odd residues, so $\Phi_g = \Psi$ on odd integers. Since $f(x) = T^{e(x)}(x)$, the difference $\Phi_g(f(x)) - \Phi_g(x)$ telescopes over the $e(x)$ steps of T, and a failure at x forces $\Psi(T(y)) > \Psi(y)$ for some $y = T^i(x)$, $0 \leq i < e(x)$. By Lemma 8.3 this requires either $k(y)\log_2 3 > m$, where $k(y)$ counts the letters $\mathtt{O}$ among the first m letters of y, or (y odd and) $\varepsilon(y) > (m - k(y)\log_2 3)/m > 0$; as $\log_2 3$ is irrational the second alternative confines y to a finite set S, and $x = (2^i y - 1)/3$ to a finite set. Let $B = \{y : k(y)\log_2 3 > m\}$, a union of classes modulo 2^m, and $G_i = \{x \text{ odd} : e(x) > i,\ T^i(x) \in B\}$. Then $F(g) \subseteq \bigcup_{i\geq 0} G_i \cup (\text{finite})$, and the union has natural density $\sum_i d(G_i)$ (the tail $\bigcup_{i>I} G_i$ lies in $\{e(x) > I\}$, of density 2^{-I}).

For $i \geq 1$ and $x \in G_i$ the window of $T^i(x)$ is $p_i \cdots p_{i+m-1}$ with $p_i = \mathtt{E}$, and by Lemma 8.1, conditionally on $p_1 = \cdots = p_i = \mathtt{E}$ (an event of density 2^{-i}) the letters $p_{i+1}, \ldots, p_{i+m-1}$ are independent fair bits. Hence $d(G_i) = 2^{-i}\Pr[\mathrm{Bin}(m-1, \frac{1}{2}) > \theta m]$, and likewise $d(G_0) = \Pr[1 + \mathrm{Bin}(m-1, \frac{1}{2}) > \theta m]$. Both are at most $\Pr[\mathrm{Bin}(m, \frac{1}{2}) > \theta m - 1]$, so

$$d\Big(\bigcup_i G_i\Big) \leq 2\Pr\big[\mathrm{Bin}(m, \tfrac{1}{2}) > \theta m - 1\big] \leq 2 \cdot 2^{-m(1-H(\theta-1/m))} \leq 2 \cdot 2^{-cm}\, 2^{|H'(\theta)|} < 4 \cdot 2^{-cm},$$

using the entropy bound (valid as $\theta m - 1 \geq m/2$ for $m \geq 8$), concavity of H, and $|H'(\theta)| = \log_2 \frac{\theta}{1-\theta} = 0.77\ldots$. □

Local coordinates of unbounded depth

The fuel τ and the detectors λ_d read x to unbounded 2-adic depth, but only along finitely many 2-adic directions. Call a coordinate α on odd integers (m, E)-*local*, for a finite $E \subset \mathbb{Z}_2$, if $\alpha(x) = \alpha(x')$ whenever $x \equiv x' \pmod{2^m}$ and $x \not\equiv t \pmod{2^m}$ for all $t \in E$.

Lemma 8.4. *τ is $(m, \{-1\})$-local for every m, and λ_d is $(m, \{t_d\})$-local for every m, where $t_d = 1 - 2^{2d+1}3^{-d} \in \mathbb{Z}_2$ is the 2-adic limit of the level-d templates.*

Proof. $\tau(x) \geq m$ iff $x \equiv -1 \pmod{2^m}$, and $\tau(x) < m$ is determined by $x \bmod 2^m$. For the detector, $T_s^{(d)} - t_d = 2^{2d+1+\mathrm{ord}_d\, s}3^{-d}$ has 2-adic valuation $2d + 1 + \mathrm{ord}_d\, s$, while $T_s^{(d)} < 2^{2d+1+\mathrm{ord}_d\, s}/3^d$ gives $\beta_d(s) \leq 2d + \mathrm{ord}_d\, s$; so $T_s^{(d)} \equiv t_d \pmod{2^{\beta_d(s)+1}}$. A match $x \equiv T_s^{(d)} \pmod{2^{\beta_d(s)+1}}$ with $\beta_d(s) + 1 > m$ therefore forces $x \equiv t_d \pmod{2^m}$; away from that class, $\lambda_d(x)$ is decided by the templates with $\beta_d(s) + 1 \leq m$, hence by $x \bmod 2^m$. □

Proposition 8.5. *Let α be (m, E)-local and let $m \geq 12$ satisfy $32(m+1)^2(m+2)|E| \leq 2^{(m+1)H(\theta)}$. Then for every g the failure set of $\log_2 x + g\big(\alpha(x), x \bmod 2^m\big)$ contains a union of residue classes of density at least $2^{-cm}/(32(m+2)^5)$. In particular this holds for $g\big(x \bmod 2^m, \tau(x), \lambda_1(x), \ldots, \lambda_D(x)\big)$ with $|E| = D+1$, while the upper bound of Theorem 8.2 applies verbatim.*

Proof. Run the proof of Theorem 8.2(1) with the coordinate $(\alpha(x), x \bmod 2^m)$. The argument of Steps 2–3 needs only that α is constant on each A_i and on $\{f(x) : x \in A_i\}$, which holds as soon as no cycle member y_i is congruent modulo 2^m to a point of E. A rotation class $[w]$ violating this has an m-window equal to the window of some $t \in E$; the window occupies all but one of the $P = m+1$ cyclic positions, so at most $2P|E|$ classes are lost. The hypothesis makes $2P|E| \leq |\mathcal{C}|/2$, and the bound follows with the constant halved. □

Remark 8.6. (i) The exponent $c = 1 - H(1/\log_2 3)$ is the rate at which the density of integers whose orbit has not descended after m parity steps decays, the event $k > \theta m$ of the proof; the same event controls the density of integers whose stopping time exceeds m (Terras [13]; see [6]). Theorem 8.2 says that a correction which may consult m bits of x cannot fail less often than the orbit itself fails to descend within those bits, and that plain lookahead attains this. (ii) The bounds are sharp only in the exponent. Exact values of the two sides for small m are tabulated in the repository; for $m \leq 13$ the lower-bound sum over expanding cycles is between $2^{-7.2}$ and $2^{-2.4}$, far above the stated bound, and the lookahead density is $2^{-1.8}$ at $m = 8$. The polynomial factors are an artefact of the proof. (iii) The lower bound for Archimedean coordinates (length, the quantized logarithm) would require the multi-witness cycles of Section 7 at every scale; we expect the same exponent for fixed precision and have not carried this out. (iv) The proof uses only Lemma 8.1 and Lemma 7.5, so it transfers to the maps $(ax+b)/c^{\mathrm{v}_c(ax+b)}$ of Remark 3.3, with the threshold $1/\log_c a$ and the corresponding binomial large-deviation rate, provided $\log_c a$ lies below the mean valuation $c/(c-1)$; above it the lookahead potential fails on a set of density tending to 1.

9 Certificates from raw orbits

The constraint system "Φ nonincreasing", restricted to any coordinate window, is a system of difference constraints $g(v) - g(u) \leq -\log_2(f(x)/x)$ over observed steps with $u = \mathrm{coords}(x)$, $v = \mathrm{coords}(f(x))$; it is infeasible iff the coordinate graph contains a directed cycle whose product of value ratios exceeds 1, and any such cycle is verifiable in exact rational arithmetic. Mining by Bellman–Ford over odd $x \leq 10^6$ yields short certificates in every window we tested: the pair $57 \mapsto 43$, $27 \mapsto 41$ closes a 2-cycle in $(x \bmod 16, \tau)$-space with product $\frac{1763}{1539} > 1$, refuting Theorem 3.2's class at $m \leq 4$ in two lines; with the level-1 detector added the miner returns $57 \mapsto 43$, $59 \mapsto 89$, and the 5-cycle $27, 41, 31, 15, 7$ through $41 \mapsto 31 = 2^5 - 1$ also closes. On the quantized-log windows of Section 7 with $x \bmod 16$ and τ it returns a 2-cycle at $j = 3$ ($110235 \mapsto 165353$, $155897 \mapsto 116923$) and a 7-cycle at $j = 5$, the lengths predicted by Proposition 7.9. Inspection of mined witnesses at larger moduli revealed the $-5, -7$ residue pattern of Section 3. This pipeline is logically independent of every lemma in this paper — it uses raw orbit steps only — and would have refuted the theorems had a valid correction existed on the tested windows.

10 Congruence certificates on the Mersenne spine

The preceding sections concern potentials. This one concerns a second method, case analysis on residue classes, applied to one explicit family of trajectories. By Lemma 2.3, $f^j(2^n-1) = 2^{n-j}3^j - 1$ for $0 \le j < n$. For odd $n \ge 3$ the n-th step divides $2(3^n-1)$ by $2^{1+\mathrm{v}_2(3^n-1)} = 4$ (LTE), so

$$f^n(2^n-1) = a_n := \frac{3^n-1}{2},$$

the base-3 repunit, and the orbit stays above 2^n-1 on the way. The stopping time of 2^n-1 (first passage below the start) is therefore $n+\sigma(a_n)$, where $\sigma(a_n)$ is the first passage of the orbit of a_n below 2^n-1. We know of no proof that $\sigma(a_n)$ is finite for all odd n, let alone $O(n)$. A natural route scores the orbit of a_n block by block and case-splits on residues of n. We show that such case splits are exact decision procedures on each residue class (Proposition 10.4), but that no finite collection of them can succeed (Theorem 10.7). The obstruction is again the cycle $-5 \mapsto -7 \mapsto -5$.

Blocks and scores

Write $T(x) = x/2$ for even x and $T(x) = (3x+1)/2$ for odd x, and read the orbit of $x \in \mathbb{Z}_2$ under T as a parity word in the letters `O` (odd) and `E` (even).

Definition 10.1. Let x be odd with $\tau(x) = 1$, so $e = \mathrm{v}_2(3x+1) \ge 2$, and let $y = f(x)$. The *block* at x consists of the step $x \mapsto y$ followed by the $\tau(y)-1$ steps of Lemma 2.3, which end at the next state with $\tau = 1$. Its *score* is

$$\Delta = 11(e-1) - 9\,\tau(y).$$

In the parity word a block reads $\mathtt{O}\,\mathtt{E}^{e-1}\,\mathtt{O}^{\tau(y)-1}$. It has $e-1$ even and $\tau(y)$ odd letters, so $\Delta = 11\cdot\#\mathtt{E} - 9\cdot\#\mathtt{O}$. A run of blocks with nonnegative total score has at most $11/20$ of its letters odd. Since $3^{11} < 2^{20}$, such a run contracts. Every block begins with the factor `OE`, and `OE` occurs nowhere else. Hence:

Lemma 10.2. *The scores $\Delta_1, \dots, \Delta_B$ of the first B blocks from x are determined by any prefix of the parity word of x that contains the first $B+1$ occurrences of* `OE`.

We use the subfamily $n = 64k + 17$. Here $\mathrm{v}_2(3^{n+1}-1) = 3$ (LTE), so $f(a_n) = x_1 := (3^{n+1}-1)/8$. Since $3^{18} \equiv 9 \pmod{32}$ and $3^{64} \equiv 1 \pmod{256}$, $x_1 \equiv 1 \pmod 4$ and $\tau(x_1) = 1$. Let $\Delta_1, \Delta_2, \dots$ be the scores of the successive blocks from x_1. For odd k one has $\Delta_1 = -16$ (Proposition 10.4 applied to the class $k \equiv 1 \pmod 2$). The score programme asks for $\Delta_2 + \Delta_3 + \cdots \ge 16$ summed over a window of $6n$ shortcut steps. We call a class of k *closed at B* if $\Delta_2 + \cdots + \Delta_B \ge 16$ on the whole class. This prefix condition does not imply the window condition, since later blocks score $-7, -16, \dots$ whenever $\tau(y) \ge 2$. Theorem 10.7 shows that even the prefix condition cannot be certified on all odd k.

Exact decision on a residue class

Lemma 10.3. *For $P \ge 6$, $x_1 \bmod 2^P$ depends only on $k \bmod 2^{P-5}$. Moreover, if $x \equiv x' \pmod{2^p}$ in $\mathbb{Z}_2$, then the parity words of x and x' agree in their first p letters.*

Proof. The order of 3 modulo 2^{P+3} is 2^{P+1}, so $3^{64k+18} \bmod 2^{P+3}$ depends on $64k \bmod 2^{P+1}$. The second statement is Terras's observation [13]: $T^i(x) \equiv T^i(x') \pmod{2^{p-i}}$ for $i \le p$, by induction. □

Proposition 10.4 (exact class decision)**.** *Fix a class $k \equiv r \pmod{2^j}$. The first $j+5$ letters of the parity word of x_1 are the same for every k in the class, including every 2-adic k. If they contain $B+1$ occurrences of* `OE`*, then $(\Delta_1, \dots, \Delta_B)$ is constant on the class. Otherwise split the class into its two children modulo 2^{j+1} and repeat. A statement about $(\Delta_1, \dots, \Delta_B)$ on a class is proved when it holds on every leaf of this tree, and refuted, with an explicit integer counterexample, when it fails on some leaf.*

Proof. Lemmas 10.3 and 10.2. □

For example, single class evaluations give

k class	$(\Delta_2, \Delta_3, \Delta_4)$	sum
$3 \bmod 2^8$	$(2, 13, 2)$	17
$171 \bmod 2^8$	$(2, 2, 13)$	17
$323 \bmod 2^9$	$(2, 13, 13)$	28
$579 \bmod 2^{10}$	$(2, 13, 4)$	19

so these four classes are closed at $B = 4$.

Remark 10.5. A single representative is not a certificate. The integers $k = 267$ and $k = 41227$ lie in the same class modulo 2^{13}. For $k = 267$, $\Delta_2 + \cdots + \Delta_6 = 43$. For $k = 41227$ the same sum is 7. The class modulo 2^{13} does not determine Δ_6. Proposition 10.4 refines it, and the leaf containing 41227 has modulus 2^{18}.

The cycle defeats every finite certificate

Lemma 10.6. *The map $k \mapsto x_1(k)$ extends to a homeomorphism from the odd 2-adic integers onto the ball $9 + 64\mathbb{Z}_2$.*

Proof. Write $3^{64} = 1 + 2^8 u$ with u odd. Then $z \mapsto (1+2^8 u)^z$ is a homeomorphism from $\mathbb{Z}_2$ onto $1 + 2^8\mathbb{Z}_2$ that scales distances by exactly 2^{-8}, and odd z go onto $257 + 2^9\mathbb{Z}_2$. The affine map $w \mapsto (3^{18}w - 1)/8$ takes $257 + 2^9\mathbb{Z}_2$ onto $(3^{18} \cdot 257 - 1)/8 + 64\mathbb{Z}_2 = 9 + 64\mathbb{Z}_2$, using $3^{18} \cdot 257 \equiv 73 \pmod{512}$. □

So every point of $9 + 64\mathbb{Z}_2$ is x_1 of some odd 2-adic k. One of them falls straight into the expanding cycle of Section 3.

Theorem 10.7 (shadow of the -5 cycle on the spine)**.** *Let $k^* \in \mathbb{Z}_2$ be defined by $3^{64k^*+20} = -367$. Then*

1. *k^* is odd, $x_1(k^*) = -47/9$, and its blocks have $\Delta_1 = -16$ and $\Delta_j = -7$ for every $j \ge 2$;*
2. *for every $J \ge 1$ and every integer $k \ge 0$ with $k \equiv k^* \pmod{2^J}$,*

$$\Delta_2 = \Delta_3 = \cdots = \Delta_B = -7, \qquad B = \left\lfloor \frac{J+2}{3} \right\rfloor;$$

3. *no finite set of residue classes of k, each closed at some B by Proposition 10.4, contains every odd integer k.*

Proof. (1) One has $-367 \cdot 3^{-20} \equiv 257 \pmod{512}$, so k^* exists and is odd by the proof of Lemma 10.6. Then $x_1 = (3^{64k^*+18} - 1)/8 = (-367/9 - 1)/8 = -47/9$. Next $3x_1 + 1 = -44/3$, so $e = 2$ and $y_1 = -11/3$ with $\tau(y_1) = \mathrm{v}_2(-8/3) = 3$, giving $\Delta_1 = 11 - 27 = -16$. Two steps of Lemma 2.3 give $-11/3 \mapsto -5 \mapsto -7$. From -7 every block is $-7 \mapsto -5 \mapsto -7$ with $e = 2$ and $\tau(-5) = 2$, so $\Delta = 11 - 18 = -7$.

(2) The parity word of $-47/9$ is $\mathtt{OEOO}(\mathtt{OEO})^\infty$. Its $(B+1)$-st occurrence of $\mathtt{OE}$ ends at letter $3B+3$. If $k \equiv k^* \pmod{2^J}$, Lemma 10.3 gives agreement in the first $J+5$ letters, which contain $B+1$ occurrences of $\mathtt{OE}$ when $3B + 3 \le J + 5$. Lemma 10.2 finishes.

(3) Suppose classes $C_1, \dots, C_m$ cover the odd integers. Refine them to a common modulus 2^J. Every odd residue modulo 2^J is then covered, including that of k^*, so some C_i contains k^*. The certificate for C_i uses only letters common to every 2-adic member of C_i (Proposition 10.4), so it applies to k^* and gives $\Delta_2 + \cdots + \Delta_{B_i} \ge 16$. By (1) that sum is $-7(B_i - 1) < 16$, a contradiction. □

Remark 10.8 (relation to Terras). Terras [13] showed that first passage below the starting point within p steps is a property of $x \bmod 2^p$, and the burn of Lemma 2.3 shows that the class $x \equiv -1 \pmod{2^p}$ has not stopped by step p, so no finite modulus decides the stopping time of all integers. Theorem 10.7 is the corresponding statement for the *index* n of the thin family $2^n - 1$, where residue classes of n are a different and a priori more promising partition; it identifies the obstruction as one explicit 2-adic index and one explicit rational cycle, and it concerns block scores, which are finer than stopping.

The mechanism is the one behind Theorem 3.2. The cycle block $-7 \mapsto -5 \mapsto -7$ has one even and two odd letters and value ratio $3^2/2^3 = \frac{9}{8} > 1$. Any linear score whose nonnegativity forces contraction must therefore score this block negatively. Residue classes of k cannot separate k from k^*, just as residues of x cannot separate x from its shadow. In (2), each extra 3 bits of modulus buy exactly one more block of shadowing. One checks that B is sharp for $J \le 120$.

How far certificates reach

Proposition 10.4 also measures the method. Among odd k, the Haar measure of the classes modulo 2^J that are closed within ten blocks is

$$\frac{1013}{2048} \approx 0.495 \ (J = 14), \qquad \frac{77177}{131072} \approx 0.589 \ (J = 18), \qquad \frac{10687}{16384} \approx 0.652 \ (J = 22).$$

The rest is undetermined at that modulus, apart from a part of measure about 2.5×10^{-4} at $J = 22$ that provably does not close within ten blocks. Each four extra bits close about 6% more. By Theorem 10.7(3) the measure never reaches 1 at any finite J.

Remark 10.9. Even a complete cover would yield only prefix inequalities. The window condition also needs the later blocks, and a lower bound for them is the same problem for the shifted orbit. The full score programme of the repository accompanying this paper, if completed, would give $\sigma(a_n) \le 6n$ and hence a stopping time of at most $7n$ odd steps for every $2^n - 1$ with n odd. That is a linear stopping-time bound for an explicit infinite family whose orbits first climb by a factor of about $(3/2)^n$, and we know of no such result.

11 The boundary

Theorems 3.2, 4.1 and 7.6, with Corollaries 7.10 and 7.11, exclude every correction of the form

$$g\big(x \bmod N,\ \tau(x),\ \lambda_1(x),\dots,\lambda_d(x),\ \lfloor 2^j \log_2 x\rfloor\big)$$
$$\text{or}\quad g\big(x \bmod N,\ \tau(x),\ \lambda_1(x),\dots,\lambda_d(x),\ \mathrm{LD}_j(x)\big),$$

for all N, d, j, along f^k for every fixed k. Every exclusion is witnessed by a finite closed cycle in a coordinate graph, every cycle is a shadow of an expanding rational cycle, and the one quantitative invariant that appears — the number of witnesses needed at precision 2^j — is the Diophantine cost of approximating $\log_2 3$ from below. Two things are not covered.

Question 11.1. Is there a nonincreasing potential $\log_2 x + g\big(\lfloor 2^j \log_2 x\rfloor,\ \mathrm{LD}_{j'}(x),\ \tau(x),\ x \bmod 2^m\big)$ for some j, j', m? The two size coordinates have breakpoints at $2^{k/2^j}$ and at $\ell\, 2^{\mathrm{len}(x)-j'}$, which do not nest, and the construction of Theorem 7.6 fixes the position of each witness in its cell only up to the width of a sub-interval. We expect a rotation argument to handle the joint coordinate, but have not carried it out. More generally one may ask the same for any finite partition of the mantissa $x/2^{\mathrm{len}(x)-1} \in [1, 2)$.

Question 11.2. Do path-dependent accountings — functions of the orbit segment leading to x rather than of x — admit analogous obstructions? Every coordinate treated here is a function of x alone, and the certificates exploit exactly that: a witness may be chosen afresh at every junction. A correction that reads how x was reached is a different object, and as the memory grows the existence of a monotone potential becomes conjecture-adjacent.

We also note what the results do *not* say. Apart from the negative statements of Section 10 about one method on one family, nothing here bears on individual trajectories, densities of divergent orbits, or cycles. The theorems delimit a proof *method*, not the dynamics.

12 Computational verification

Every numbered claim is verified by scripts using exact integer and rational arithmetic (no floating point in any assertion): the shadow lemma over wide parameter ranges with detector evaluation at all nodes (`verify_shadow.py`); the iterated shadow, the concrete Dirichlet instance $(m, j, a) = (6, 4, 53)$ at depth $N = 177$, and the irrationality arithmetic (`verify_leading_digit.py`); the tower in all parts, including both directions of the domain criterion and the carry-free normal form (`verify_tower.py`); the family-method subclasses (`verify_no_local_potential.py`, `verify_nlp2.py`); the independent certificate miner with exact cycle confirmation (`verify_nogo_certificate.py`, `verify_fuel_fraction.py`); for Section 7, the rational-cycle lemma over all 21 844 words with $K \le 7$, $e_i \le 4$, the construction of Theorem 7.6 on a grid of (j, K, N, d) with $j \le 8$ (closure at every junction, detectors, arbitrary moduli including $N = 2^5 \cdot 3^2 \cdot 5 \cdot 7$ for $j \le 6$, and the exact product gain), the admissibility table, the modulus and multi-step corollaries, and the explicit instances of Remark 7.7 (`verify_quantized_log_theorem.py`); for Section 8, the Terras bijection to $n = 14$, the lookahead identity as an exact identity in the formal symbol $\log_2 3$ for $m \le 12$, the exact failure density of the lookahead potential against $4 \cdot 2^{-cm}$ for $m \le 40$ together with a direct check of all its failures below 2^{17}, and for $m \le 13$ every expanding cyclic word of length $m + 1$ with its cycle, edge classes, gain and window, giving the exact lower-bound sums against the stated constant (`verify_failure_density.py`); and for Section 10, the ball lemma, the shadow index k^*,

part (2) of Theorem 10.7 for $J \le 120$, the four class evaluations, and the exact closure measures (`verify_residue_certificate_nogo.py --measure-22`), with the class decision procedure in `prove_block8_17_classes.py`. The scripts accompany the repository.

References

[1] P. Andaloro, *On total stopping times under* $3x+1$ *iteration*, Fibonacci Quart. **38** (2000), 73–78.

[2] D. Barina, *Convergence verification of the Collatz problem*, J. Supercomputing **77** (2021), 2681–2688.

[3] C. J. Everett, *Iteration of the number-theoretic function* $f(2n) = n$, $f(2n+1) = 3n+2$, Adv. Math. **25** (1977), 42–45.

[4] A. Ya. Khinchin, *Continued Fractions*, University of Chicago Press, Chicago, 1964.

[5] I. Krasikov and J. C. Lagarias, *Bounds for the* $3x+1$ *problem using difference inequalities*, Acta Arith. **109** (2003), 237–258.

[6] J. C. Lagarias, *The* $3x+1$ *problem and its generalizations*, Amer. Math. Monthly **92** (1985), 3–23.

[7] J. C. Lagarias (ed.), *The Ultimate Challenge: The* $3x+1$ *Problem*, Amer. Math. Soc., Providence, RI, 2010.

[8] J. C. Lagarias, *The* $3x+1$ *problem: an annotated bibliography (1963–1999)*, arXiv:math/0309224.

[9] J. C. Lagarias, *The* $3x+1$ *problem: an annotated bibliography, II (2000–2009)*, arXiv:math/0608208.

[10] J. Simons and B. de Weger, *Theoretical and computational bounds for* m*-cycles of the* $3n+1$ *problem*, Acta Arith. **117** (2005), 51–70.

[11] R. P. Steiner, *A theorem on the Syracuse problem*, Proc. 7th Manitoba Conference on Numerical Mathematics, 1977, 553–559.

[12] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum Math. Pi **10** (2022), e12.

[13] R. Terras, *A stopping time problem on the positive integers*, Acta Arith. **30** (1976), 241–252.

[14] G. J. Wirsching, *The Dynamical System Generated by the* $3n+1$ *Function*, Lecture Notes in Math. 1681, Springer, Berlin, 1998.